GURU99

Banking system

TEST PLAN

V.01

Prepared by : Areej F. Alashi

1. Introduction

This document presents the Test Plan for the Guru99 Banking System. It outlines the overall testing approach, scope, objectives, resources, and schedule required to ensure the quality of the application. The purpose of this Test Plan is to define the testing strategy and activities needed to validate that the system meets the functional and integration requirements specified in the Software Requirements Specification (SRS). This document serves as a reference for all testing activities and ensures a structured and consistent approach throughout the testing lifecycle.

2. Scope

This test plan covers the functional and integration testing of the Guru99 Banking System based on the defined requirements in the SRS.

In Scope:

The following modules will be tested to ensure correct functionality and proper interaction between system components:

- **Customer Management:**

  New Customer , Edit Customer, Delete Customer

- **Account Management:**

  New Account ,Edit Account_ Delete Account

- **Transaction Management:**

  Deposit , Withdrawal , Fund Transfer

- **Inquiry and Reporting:**

  Balance Enquiry ,Mini Statement, Customized Statement

- **Security and Access:**

Login and Logout , Change Password

- Testing will be conducted for both Manager and Customer roles, ensuring proper authorization and access control across all applicable modules.
- The scope also includes validation of data flow between modules, such as customer creation, account creation, and transaction processing.

Out of Scope:

- Non-functional testing (performance, load, and stress testing)

- Security testing beyond basic validation

- Automation testing (planned for future phases)

- Cross-browser testing beyond Chrome

3. Test Strategy and methodology

Test Strategy:

The testing strategy for the Guru99 Banking System is based on a structured approach to ensure complete validation of system functionality, data integrity, and module interactions.

Testing will be conducted at multiple levels to ensure quality across the application:

- **Functional Testing:**

To validate that all system functionalities behave according to the defined requirements in the SRS.

- **Integration Testing:**

To verify the data flow and interaction between interconnected modules such as Customer Management, Account Management, and Transaction Processing.

- **System Testing:**

To validate the end-to-end business workflows across the system.

- **User Acceptance Testing (UAT):**

To confirm that the system meets business needs and user expectations.

Methodology:

- **Requirement-Based Testing:**

All test scenarios and test cases are derived from the SRS to ensure full coverage of requirements.

- **Manual Testing Approach:**

Testing will be performed manually to simulate real-world user interactions and validate system behavior.

- **Defect Management:**

All defects will be logged, tracked, and managed using a defect tracking tool. Each defect will be assigned severity and priority levels and will go through a defined lifecycle until closure.

- **Test Case Design:**

Test cases will be designed using standard techniques such as equivalence partitioning, boundary value analysis, and error guessing.

- **Test Execution:**

Test cases will be executed systematically, and results will be documented to ensure traceability and coverage.

- **Future Automation:**

Critical and repetitive test scenarios will be identified for future automation using tools such as Selenium.

4. Test Environment:

- The application is compatible with Google Chrome version 27 and above.
- A stable and uninterrupted internet connection is required.
- A standard computer/laptop will be used for testing.
- Currently, the application is web-based and does not support mobile or other browser versions.
- Testing will be performed in a controlled environment to ensure consistent and reliable results.

5. Entry and Exit Criteria:

Entry Criteria:

- Business requirements are clearly defined, documented, and approved by all stakeholders.
- The Software Requirements Specification (SRS) is complete, stable, and baselined for testing.
- All required test scenarios and test cases are designed, reviewed, and signed off.
- The test environment is fully set up, configured, and validated for execution.
- Required test data is available and prepared.
- The application build is deployed and smoke-tested successfully.

Exit Criteria:

- All planned test cases have been executed and major functionalities are tested with acceptable pass rate.
- All critical and high-severity defects are resolved, retested, and closed.
- No open blocker or critical defects remain in the system.
- End-to-end business workflows have been successfully validated across all integrated modules.
- Test coverage requirements have been met as per the defined scope in the Test Plan.
- Test results have been documented and reviewed.
- Test Summary Report has been prepared, reviewed, and shared with stakeholders for approval.

6. Test Deliverables :

- Test Plan document
- Test Scenarios and Test Cases document
- Defect/Bug Report
- Test Execution Report
- Test Summary Report

7. Risks & Assumptions

Risks:

- QA environment may become unavailable or unstable during test execution.
- Application may crash or behave unexpectedly during testing.
- Limited time before release may result in incomplete test execution.
- Unplanned team leaves may affect testing progress.
- Lack of required technical skills may slow down testing activities.
- Developers may not be responsive, causing delays in defect resolution.
- Misunderstanding of requirements or test cases may lead to incorrect test execution.

Assumptions:

- The application under test is stable and accessible during the testing phase.
- Required resources (environment, tools, and test data) are available.
- SRS is complete and not frequently changing during execution.

8. Test Schedule

- Test Planning: 1–2 days
- Test Design (Scenarios & Test Cases): 2–4 days
- Test Environment Setup & Smoke Testing: 1–2 days
- Test Execution: 5–10 days (depending on scope)
- Defect Retesting & Regression Testing: 3–5 days
- Test Closure & Reporting: 1–2 days